Alternative Energy ETF Portfolio

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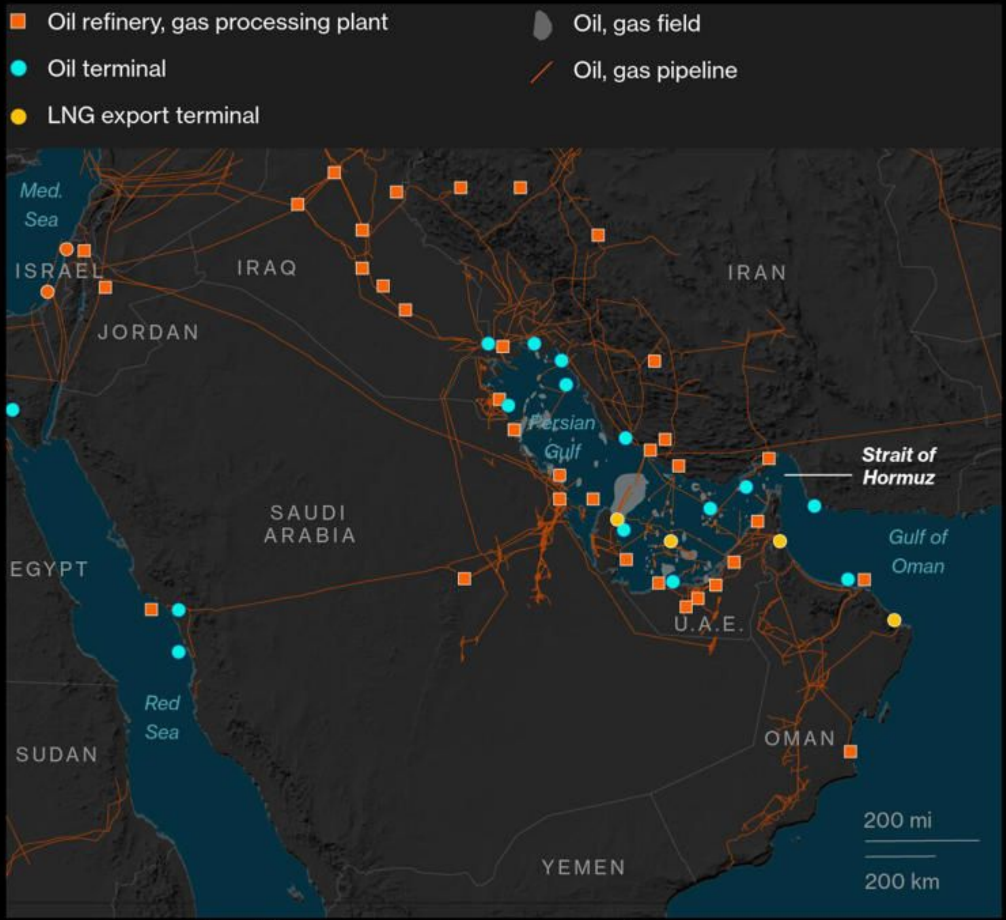
Oil Refineries and Terminals in the Middle East

About a fifth of the world's oil passes through the Strait of Hormuz



Source: US Central Intelligence Agency, US Department of Energy

Bloomberg

30%	Global seaborne oil trade
20%	Global liquified natural gas trade
20 million	Barrels per day of crude oil and petroleum products

Equally Weighted Benchmark Portfolio

TAN

Solar Energy ETF

Global solar energy manufacturing, equipment, development, infrastructure, and financing companies

- ➔ Lowest ongoing international dependence
- ➔ Commodity risk
- ➔ Policy risk
- ➔ Acts like a tech growth stock

FAN

Wind Energy ETF

Wind turbine manufacturers, wind farm developers & utilities, wind supply chain and infrastructure companies

- ➔ Dependencies on China for solar manufacturing and raw materials
- ➔ Infrastructure risk
- ➔ Interest rate risk
- ➔ Acts like an infrastructure stock

LIT

Lithium Battery ETF

Lithium miners and raw materials, refining and chemical processing, battery manufacturers, supply chain, and equipment

- ➔ Dependencies on Australia, Chile, China, and Democratic Republic of Congo
- ➔ Swaps oil dependence for mineral dependence
- ➔ Acts like a growth commodity

ICLN

Blended Alternative Energy ETF

Utility and infrastructure heavy alternative energy stocks with global exposure.

- ➔ More stable cash flow businesses
- ➔ Supply chain risk
- ➔ Policy risk
- ➔ Market cycle risk
- ➔ Acts like an infrastructure/utility stock

QCLN

US-Centric Alternative Energy ETF

More American alternative energy tech, electric vehicle, and semiconductor exposure (tech/growth ETF with an alternative energy theme)

- ➔ Driven by innovation and multiples
- ➔ High interest rate sensitivity
- ➔ Higher volatility than ICLN
- ➔ Acts like a tech growth stock

Active portfolio strategy: Tilt Exposures

Additional risky assets for the active portfolio

SPY

S&P 500 ETF

Proxy for overall market beta

Allows optimizer to separate sector vs market exposure to hedge broad market drawdowns

QQQ

Growth Tech ETF

Proxy for Growth/Duration Risk

Captures sensitivity to interest rates and valuation multiples

Adjust exposure to tech-driven market cycles

TLT

Long duration US Treasuries

Proxy for Interest Rate Risk

Offsets rate sensitivity

Shorting TLT positions the portfolio for rising rate environments

Portfolio Optimization

Objective

Maximize risk-adjusted return (mean-variance) and improve it relative to the benchmark

Higher information ratio

Portfolio structure

Active portfolio around equal-weight benchmark

Reallocates risk with the goal of improving risk-adjusted return

Assets

Core: Alternative energy ETFs

Hedge: SPY, QQQ, TLT

Benchmark → + Active tilt → Final portfolio

Baseline Optimization

·No constraints on weights
·Allows large long/short positions
·Highly sensitive to estimated returns

	Average Weight (%)	Min Weight (%)	Max Weight (%)
ICLN	-13.67	-30.00	70.00
TAN	30.28	-30.00	70.00
FAN	17.02	-30.00	70.00
LIT	11.27	-30.00	70.00
QCLN	25.25	-30.00	70.00
SPY	19.89	-50.00	50.00
QQQ	45.91	-24.48	50.00
TLT	-35.96	-50.00	31.54

Extreme weights observed
Longest on QQQ (Growth Tech), TAN (Solar), and QCLN (Clean energy tilted Tech ETF)
Short ICLN (Blended clean energy ETF) and TLT (Long duration treasuries)

- Small differences in estimated returns (μ) drive large weight changes
- These estimates are highly **noisy and unstable**
- The optimizer treats noise as signal
- Result: extreme positions based on **unreliable inputs**

Optimizer is leveraging estimation error instead of finding alpha

Stabilizing the Portfolio with Constraints

Core ETFs constrained to remain positive
Hedge ETFs allowed to be short
Weight caps introduced (≈30%)

	Average Weight (%)	Min Weight (%)	Max Weight (%)
ICLN	11.00	5.00	35.0
TAN	20.14	5.00	35.0
FAN	20.24	5.00	35.0
LIT	18.00	5.00	35.0
QCLN	24.88	5.00	35.0
SPY	4.95	-15.00	15.0
QQQ	12.60	-14.63	15.0
TLT	-11.81	-15.00	15.0

	Basic Optimization	Constrained Optimization
Annualized Return	0.122834	0.129144
Annualized Volatility	0.361163	0.293626
Sharpe Ratio	0.340107	0.439823
Max Drawdown	-0.730132	-0.622852

	Equal-Weight Benchmark
Annualized Return	0.137794
Annualized Volatility	0.254932
Sharpe Ratio	0.540513
Max Drawdown	-0.547463

Constraints added to prevent concentration risk, reduce turnover, and promote factor hedge behavior of added ETFs

Constraints significantly reduce extreme tilting and improve stability

Improving Input Estimation

We tested alternative input estimation methods to reduce estimation error. Shrinking expected returns improves performance, suggesting that return estimates are highly noisy. **Optimization is μ-driven, not Σ-driven. In a small ETF universe, μ is mostly noise.** In contrast, covariance shrinkage alone does not improve results, and may even worsen performance. This indicates that estimation error in expected returns is the primary driver of instability in the optimization.

Rebalancing Frequency

	Monthly	Quarterly	Semiannual	Equal-Weight Benchmark
Annualized Return	0.134846	0.131576	0.131793	0.137794
Annualized Volatility	0.286406	0.286845	0.289301	0.254932
Sharpe Ratio	0.470822	0.458699	0.455557	0.540513
Max Drawdown	-0.599747	-0.612026	-0.614722	-0.547463

	Average Weight (%)	Min Weight (%)	Max Weight (%)
ICLN	10.06	5.00	35.00
TAN	21.48	5.00	35.00
FAN	19.63	5.00	35.00
LIT	17.84	5.00	35.00
QCLN	23.80	5.00	35.00
SPY	5.85	-15.00	15.00
QQQ	13.45	-1.27	15.00
TLT	-12.10	-15.00	12.99

Rebalancing frequency is not the key performance driver

Backtesting setup

How we test

→ Rolling 36-month training window for μ, Σ

→ Walk forward; refit each rebalance; evaluate next month realized return

→ Best variant: constrained + shrink μ + sample Σ + monthly

→ Risk-free rate from Ken French monthly RF series

→ Fama-French 6-factor regression for alpha attribution

Out-of-sample window

152

months of OOS data
Sep 2013 – Apr 2026

Universe

→ Core: ICLN, TAN, FAN, LIT, QCLN

→ Hedge: SPY, QQQ, TLT

→ Benchmark: equal-weighted core basket

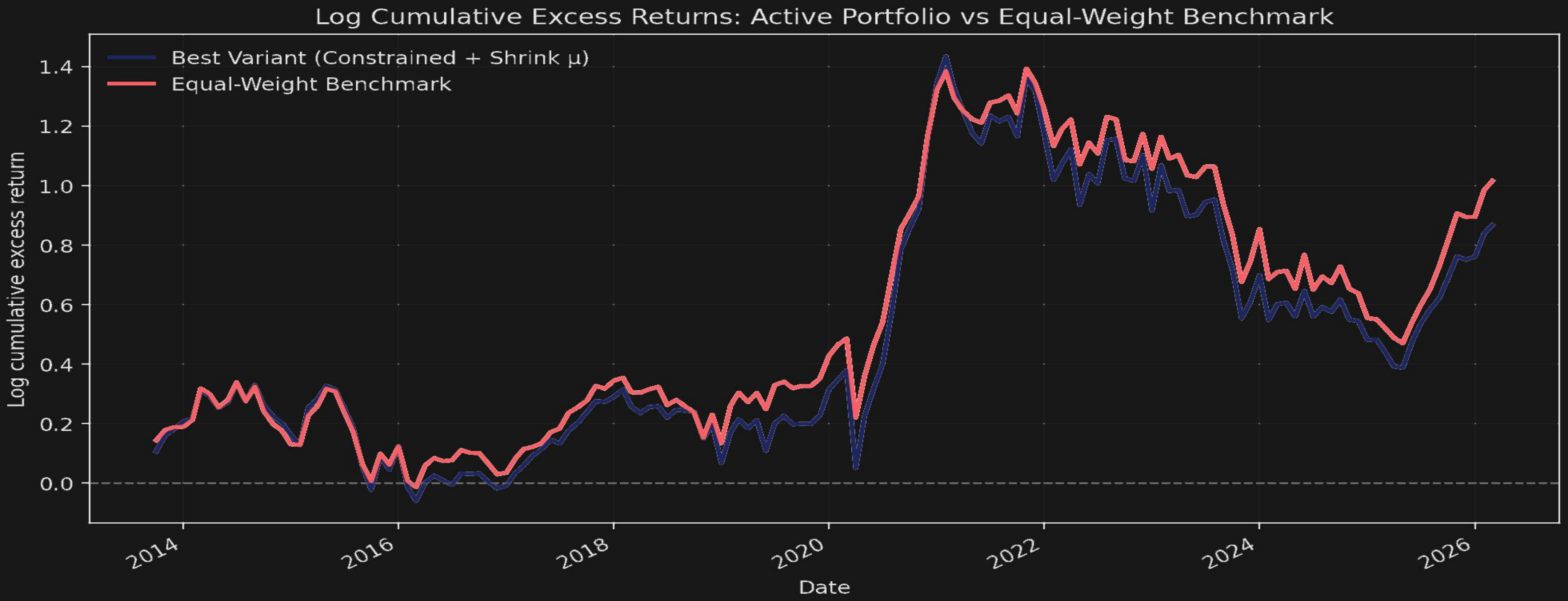
We compare the optimized active book against a simple equal-weight core basket on the same OOS window.

Out-of-sample performance

Metric	Active (Best Variant)	Equal-Weight Benchmark
Annualized excess return (vs. real RF)	11.05%	11.40%
Annualized volatility	28.70%	25.56%
Sharpe ratio	0.38	0.45
Tracking error	7.08%	-
Information ratio	-0.05	-
Max drawdown	-64.9%	-60.3%

Active book takes more risk, earns less return - both Sharpe and IR favor the simple 1/N benchmark.

Cumulative excess return: Active vs Benchmark



Active portfolio tracks consistently below benchmark – including the 2020–21 clean-energy boom.

Fama-French 6-factor decomposition

FF6 alpha (annualized)

-1.62%

Active book
t-stat = -0.29
$R^2 = 0.578$

+0.40%

Equal-weight benchmark
t-stat = 0.08
$R^2 = 0.552$

Both alphas are statistically indistinguishable from zero ($|t| << 2$). Neither portfolio earns abnormal return after factor adjustment.

Factor loadings

Factor	Active	Benchmark
Mkt-RF (β)	1.25	1.08
SMB	0.48	0.49
HML	-0.36	-0.28
RMW	-0.49	-0.38
CMA	0.07	0.09
Mom	-0.13	-0.10

The optimizer adds ~17% more market beta (1.25 vs 1.08), explaining higher volatility - but no alpha.

What we learned

1

Estimation error dominates

Even with shrinkage on μ and tight bounds, the active book underperforms 1/N. Sample-mean returns are too noisy in a 5-asset universe with monthly data.

2

No clean-energy alpha

FF6 explains ~58% of variance with zero residual α. The basket's outperformance vs broad market is market β plus small-cap tilt - not unique skill.

3

Higher β, no reward

Optimization tilts the book to $\beta = 1.25$ vs 1.08 for the benchmark. It buys more market exposure without buying alpha.

4

Possible improvements

Predictor-based μ, longer estimation windows, Black-Litterman views, or theme rotation across multiple sleeves.

1/N is hard to beat with a small universe. Negative results are still informative - they tell us where the inefficiency is not.

Future Iterations to Improve Results

Improve expected return estimates (μ)

- ➔ Use macro/factor predictors instead of sample means
- ➔ Apply forecast combination or shrinkage toward benchmark

Strengthen optimization framework

- ➔ Use active portfolio (maximize information ratio) more explicitly
- ➔ Add tighter constraints on tracking error and weight deviations

Expand Asset Universe

- ➔ Add complementary assets (e.g., broader sectors or commodities)
- ➔ Reduce correlation and estimation error

Refine Asset Universe and Hedge Structure

- ➔ Allow shorting only on hedge assets or remove entirely (SPY, QQQ, TLT)

Incorporate simple factor structure

- ➔ Use market, growth, and rate factors to guide allocation
- ➔ Avoid unintentionally increasing β without improving returns

Refine covariance estimation (Σ)

- ➔ Use shrinkage estimators (e.g., Ledoit-Wolf) or EWMA
- ➔ Improve stability of risk estimates

Conclusions

- Optimization alone is unstable
- Constraints improve robustness
- Input estimation matters most

Even with improvements, the optimized portfolio does not outperform the equal-weight benchmark